



## Re-evaluating the need for chronic toxicity studies with therapeutic monoclonal antibodies, using a weight of evidence approach

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### ABSTRACT

To support registration of monoclonal antibodies (mAbs) for chronic indications, 6-month toxicity studies have historically been conducted. Experience with mAb development has shown a relatively benign and well-understood safety profile for this class, with most toxicity findings anticipated based on pharmacology. We evaluated whether a 6-month toxicity study is necessary to assess the long-term safety of mAbs. Data on First-in-Human (FIH)-enabling and chronic toxicity studies were shared for 142 mAbs submitted by 11 companies. Opportunities to further optimize study designs to reduce animal usage were identified. For 71% of mAbs, no toxicities or no new toxicities were noted in chronic studies compared to FIH-enabling study findings. New toxicities of potential concern for human safety or that changed trial design were identified in 13.5% of cases,

**Abbreviations:** ADA, Anti-drug antibody(ies); ADCs, Antibody drug conjugates; CDR, Complementarity-determining region; CNS, Central nervous system; EPAA, European Partnership for Alternative Approaches to Animal Testing; EFPIA, European Federation of Pharmaceutical Industries and Associations; FIH, First-in-Human; GLP, Good laboratory practice; ICH, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use; MEB, Medicines Evaluation Board; M + F, Male and female; mAb, Monoclonal antibody; NC3Rs, National Centre for the Replacement, Refinement & Reduction of Animals in Research; NHP, Non-human primate; NOAEL, No-observed-adverse-effect level; PD, Pharmacodynamic; 3Rs, Replacement, Refinement & Reduction of Animals; TDAR, T-cell-dependent antibody response; TG, Transgenic; TK, Toxicokinetic.

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Nonclinical  
3Rs  
Reduction  
Refinement

with 7% being considered critical and 2% leading to program termination. An iterative, weight-of-evidence model which considers factors that influence the overall risk for a mAb to cause toxicity was developed. This model enables an evidence-based justification, suggesting when 3-month toxicity studies are likely sufficient to support late-stage clinical development and registration for some mAbs.

## 1. Introduction

Evaluation of the safety of new drugs in animals is performed to characterize toxicity and minimize progression of unacceptably toxic molecules to the clinic, to inform the selection of safe clinical starting doses for humans, and to inform clinicians on relevant measures for safety monitoring parameters and potential risk factors. These studies are guided by a comprehensive set of international guidelines, such as the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). Of these, ICH S6(R1) has been developed specifically for the nonclinical safety evaluation of biotechnology-derived products as these molecules differ substantially from conventional small molecule drugs produced by chemical synthesis and therefore have different considerations for safety assessment.

For biopharmaceutical intended for chronic use, a maximum toxicity study duration up to 6 months was generally considered appropriate, as originally cited in ICH S6 (1997). The 2011 revision to ICH S6 (ICH S6(R1), 2011) reinforced that studies of 6 months duration were considered sufficient and stated that studies longer than 6 months duration have not generally provided useful information for clinical development. For the treatment of advanced cancer, studies of 3 months duration are sufficient for marketing authorization under ICH S9 (Nonclinical Evaluation for Anticancer Pharmaceuticals) guidance (ICH S9, 2009).

The duration of studies to support nonclinical development of biopharmaceuticals has been investigated both before and after the ICH S6 revision. In 2008, Clarke et al. evaluated the findings in chronic toxicity studies for 23 biotechnology-derived products concluding that clinically relevant findings were identified in the 6-month studies, which supported continued use of this duration (Clarke et al., 2008). However, in 2012, the UK National Centre of Replacement, Refinement and Reduction of Animals in Research (NC3Rs) reported that for 33 out of 51 mAbs, no test article-related findings were noted in either shorter-term or longer-term studies (majority of which were 6 months in duration), suggesting that longer-term studies might not be needed (Chapman et al., 2012). In a 2016 workshop, industry experience indicated that for 30 out of 32 mAbs the toxicological profile determined in the 3-month toxicity studies did not differ from those in 6-month studies. When findings of toxicological relevance were observed in chronic studies, early signs of toxicity were also detected in the 3-month studies, suggesting that, on a case-by-case basis, there are opportunities to waive or to reduce the duration of the chronic studies (Blaich et al., 2016). While these previous investigations suggested that shorter study durations were sufficient in mAb safety evaluation, their small sample sizes were insufficient to conclude whether a 6-month study is still necessary.

ICH S6(R1) provides flexibility in design of the overall nonclinical program. For example, safety evaluation programs should normally include one rodent and one nonrodent species, but only one species may be used when only one pharmacologically-relevant species can be identified or where the biological activity of the biopharmaceutical is well understood (ICH S6(R1), 2011). Another example of flexibility is that chronic studies can be conducted in a single species, if the toxicity findings in shorter term studies are comparable (ICH S6(R1), 2011). The flexibility provided by ICH S6(R1) could be applied to reducing the duration of chronic toxicity studies with mAbs under appropriate circumstances. The rationale is that the toxicity of mAbs is generally restricted to exaggerated pharmacology and/or immunogenicity that may result in hypersensitivity reactions (Brennan et al., 2010; Hansel et al., 2010). Thus, the potential toxicity related to pharmacology can

often be predicted. Immunogenicity to mAbs and other therapeutic proteins in animals is not considered predictive of humans, and further, hypersensitivity reactions to therapeutic proteins are always possible and animal studies are usually not needed to understand this risk (Leach et al., 2014; Rojko et al., 2014; Kronenberg et al., 2017). Therefore, we conducted a survey to investigate whether repeat-dose toxicity studies with mAbs of  $\geq 6$ -months duration provided toxicity data informing human safety, clinical trials and registration when compared to shorter duration toxicity studies. We also determined what conditions and scientific factors should be involved in determining when 6-month studies might be necessary vs. utilizing shorter studies, and if there is a shorter duration toxicity study that could potentially support Phase 1, later clinical development, and registration for marketing. This project was initiated by the Netherlands Medicines Evaluation Board (MEB) and the European Partnership for Alternative Approaches to Animal Testing (EPAA) and was co-led by industry representatives and the NC3Rs. An industry working group was formed from EPAA-member companies and other industry partners, representing 14 global pharmaceutical and biotechnology companies, and representatives from NC3Rs, European Federation of Pharmaceutical Industries and Associations (EFPIA) and the European Commission.

## 2. Materials and methods

### 2.1. Data collection

A questionnaire was developed by the working group and administered by the NC3Rs. The survey was distributed to industry partners, and consisted of three main sections (basic information, study information, and comparison of toxicities) to obtain details on the mAb properties, key toxicities identified in each species and study, whether or not toxicities were consistent with pharmacology, and the impact of the data on development of the mAb or on clinical study designs (See [Supplementary Table 1](#) for a blank survey). In brief, the following information was requested.

#### Section 1 – general information

- Basic test article information such as modality type, IgG subtype, current development status (marketed, in development, or stopped), and the primary disease area of development.
- Details on the nonclinical species selected for toxicity studies, including selection process and justification.
- In scope: Standard, modified, or other mAb sub-types that have complementarity-determining region (CDR) binding domains.
- Out of scope: Fc-fusion proteins and homologous proteins without CDR binding domains and antibody drug conjugates (ADCs).

#### Section 2 – study information

- Information on each study performed with the mAb including study start date, species used, design (duration, dose route and regimen, number and size of main study and recovery groups), no-observed-adverse-effect level (NOAEL) and associated exposure multiple, and information on the nature of the findings defining the NOAEL.
- In our survey, findings that were not adverse and not considered a concern for human safety were assigned as ‘No findings’. New toxicities were identified when adverse findings or findings of human concern were observed in studies  $\geq 6$  months duration while no findings or findings of reduced incidence/severity were observed in shorter duration toxicity studies.

- When adverse findings or findings of human concern were observed, information was requested on the nature of these findings, i.e., if they were considered exaggerated pharmacology, immune-mediated toxicities, off-target effects or mediated via a different mechanism.
- Immune-mediated toxicities were defined as anti-drug antibody (ADA)-mediated toxicities, hypersensitivity, or unintended adverse effects on the immune system.
- Off-target effects were defined as findings (other than immune-mediated toxicities) known or considered to be unrelated to the intended primary pharmacology of the mAb.
- When the reported finding could not reasonably be attributed to one of these categories, they were listed as 'other' and an explanation provided via comments boxes, if applicable.

#### Section 3 – comparison of toxicities

- A comparison of the observed toxicities in shorter-term (FIH-enabling) and chronic studies supporting later development focused on whether new toxicities were observed in chronic studies, i.e., toxicities that had not been observed in short-term studies.
- When new toxicities were identified in chronic studies, information was requested on the impact of these new toxicities on human safety, and if they resulted in changes to the clinical study design and/or stopped clinical progression.

Survey respondents could provide additional information via free-text comment boxes throughout the survey. No proprietary information was requested or furnished (mAb identifying information such as name, alias or chemical structures, or the target).

Completed questionnaires were submitted by individual companies during the period of March to December 2020 and data were collated and anonymized by NC3Rs before being provided to the primary researcher at the MEB for analysis.

Data were collected from pivotal toxicity studies performed with mAbs that are marketed, still in active development, or whose development was terminated. The data were evaluated to identify opportunities to optimize the duration of nonclinical studies with mAbs and to determine whether chronic studies (of 6 months duration) provided additional information of toxicological relevance that informed human safety. A workshop was held in April 2021 with over 30 experts representing the pharmaceutical and veterinary industry, European health authorities and the NC3Rs to obtain consensus on the interpretation of the results.

##### 2.1.1. Data inclusion criteria

A separate survey response for each mAb was requested. Inclusion criteria for each mAb were defined as follows.

- (1) At least two Good Laboratory Practice (GLP)/pivotal studies of different duration, with at least one study supporting each phase of development (FIH-enabling or later development).
- (2) The data provided were related to studies for a single mAb.
- (3) Studies generally involved repeat-dose administration. Single-dose studies were only included when exposure and/or pharmacology was maintained over a prolonged observation period.
- (4) The same or similar dose levels were used between studies of different duration, and generally the same endpoints were evaluated.

##### 2.1.2. Definitions

The following guidance/definitions were used to assist completion of the survey.

- 'Standard mAbs' were defined as those with no Fc modifications and 'Modified mAbs' as those with one or more Fc modifications. Other

mAb sub-types such as bi/multi-specific mAbs, Fabs, and nanobodies were categorized as 'Other mAbs'.

- 'New toxicities' were defined as toxicities that were exclusively observed in chronic studies and either of human concern or not of human concern but with impact on clinical study design.
- 'Critical new toxicities' were defined as new toxicities that were exclusively observed in chronic studies that would have put patients at risk had they not been identified.
- 'ADA-mediated toxicities' were defined as findings known/considered to be related to, and mediated by, ADA formation.
- 'Hypersensitivity' was defined as exaggerated or inappropriate immune response resulting in local or general tissue injury, from type I to IV reactions.
- 'Immunotoxicity' was defined as unintended, adverse effect on the immune system, e.g., unintended immune suppression or enhancement.
- 'GLP/pivotal studies' are defined as studies performed to support FIH and chronic dosing. These are generally GLP-compliant, or occasionally GLP-like (to the greatest extent feasible).

#### 2.2. Analysis

Monoclonal antibodies were grouped for analysis according to study durations reported in the surveys. Short-term was defined as single dose up to 10 weeks duration (supporting FIH trials); sub-chronic defined as 12–16 weeks duration (supporting either FIH trials or later development); and chronic defined as 24 weeks duration or longer (supporting later clinical development). Following some survey exclusions (Fig. 1A and Section 3.2), three primary groups of mAb data sets were apparent (Fig. 1B).

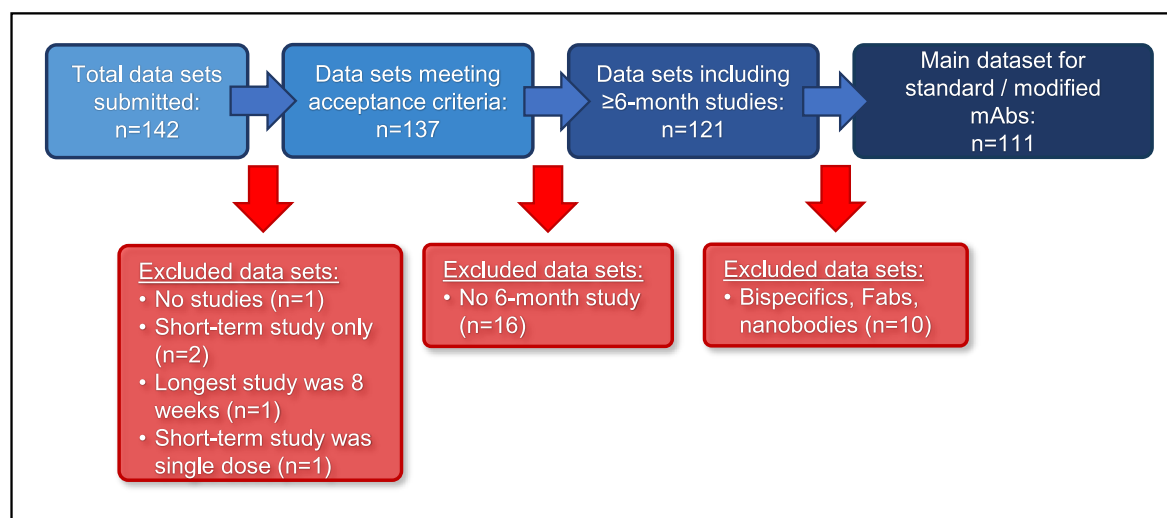
- 1) mAb data sets with short-term and chronic studies. This group was further divided into one sub-group of data sets with  $\leq 4$  weeks and  $\geq 24$  weeks studies and another sub-group of data sets with 5–10 weeks and  $\geq 24$  weeks studies weeks, for sub-analysis.
- 2) mAb data sets with sub-chronic and chronic studies.
- 3) mAb data sets with short-term, sub-chronic, and chronic studies.

The data sets were stratified for different analytical purposes and the rates of new toxicities in each of the associated groups or clusters were assessed and compared.

Firstly, the data sets were stratified into those with new toxicities vs. those without new toxicities in the chronic study. In mAb data sets with novel findings, the number of data sets with new toxicities of human concern were identified. Then, the types of new toxicities were stratified by impact i.e., whether they were anticipated and whether they impacted clinical study design or development, based on the pharmacological mode of action of the mAb. For those mAb data sets with novel findings not of concern for human safety, the types of findings and whether they could have been anticipated were also analysed.

Secondly, quantitative comparisons were made to evaluate whether specific parameters such as primary disease areas, exposure multiples, or study duration had impact on the likelihood of new toxicities occurring in the chronic studies.

Thirdly, a qualitative analysis of the reported new toxicities was carried out to evaluate how these new toxicities contributed to human risk assessment, and if they impacted the progression of the clinical program or trial design. If clinical trial subjects were deemed to be at risk of safety related events without these new toxicities having been identified, such findings were considered critical for human risk assessment and the resulting changes in clinical trial progression or design were considered meaningful. Additional questionnaires were sent out to the companies that had submitted chronic studies with new toxicities (see [Supplementary Table 2](#) for a blank survey).

**A****B**

| Complete data set                                                   | Study Duration Pairs                                                                     | New Toxicities                           |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------|
| Main data sets for standard / modified mAbs: (n=111)                |                                                                                          | <b>Of human concern</b>                  |
|                                                                     |                                                                                          | No change in clinical trial design (n=2) |
|                                                                     |                                                                                          | Change in clinical trial design (n=5)    |
|                                                                     |                                                                                          | Change in clinical trial design (n=1)    |
| Data sets with new adverse toxicity finding in chronic study (n=32) | <12 vs. ≥24 weeks duration: n=65<br>New toxicities: n=10 (15%)                           | <b>Not of human concern</b>              |
|                                                                     |                                                                                          | Change in clinical trial design (n=2)    |
| New toxicities of human concern / clinical impact (n=15)            | 12-16 weeks vs. ≥24 weeks duration: n=22<br>New toxicities: n=2 (9%)                     | <b>Of human concern</b>                  |
|                                                                     |                                                                                          | Change in clinical trial design (n=1)    |
|                                                                     |                                                                                          | Change in clinical trial design (n=1)    |
| New toxicities that were considered critical (n=8)                  | <12 weeks and/or 12-16 weeks vs. ≥24 weeks duration: n=24<br>New toxicities: n=3 (12.5%) | <b>Of human concern</b>                  |
|                                                                     |                                                                                          | Change in clinical trial design (n=2)    |
| Termination of clinical program (n=2)                               |                                                                                          | <b>Not of human concern</b>              |
|                                                                     |                                                                                          | Change in clinical trial design (n=1)    |

**Fig. 1.** Results of analysis.

**Fig. 1A.** Data inclusion and exclusion. **Fig. 1B.** Risk perception and identification of data sets with new toxicities in the respective study of longer duration.

### 3. Results

#### 3.1. Overview of the data set

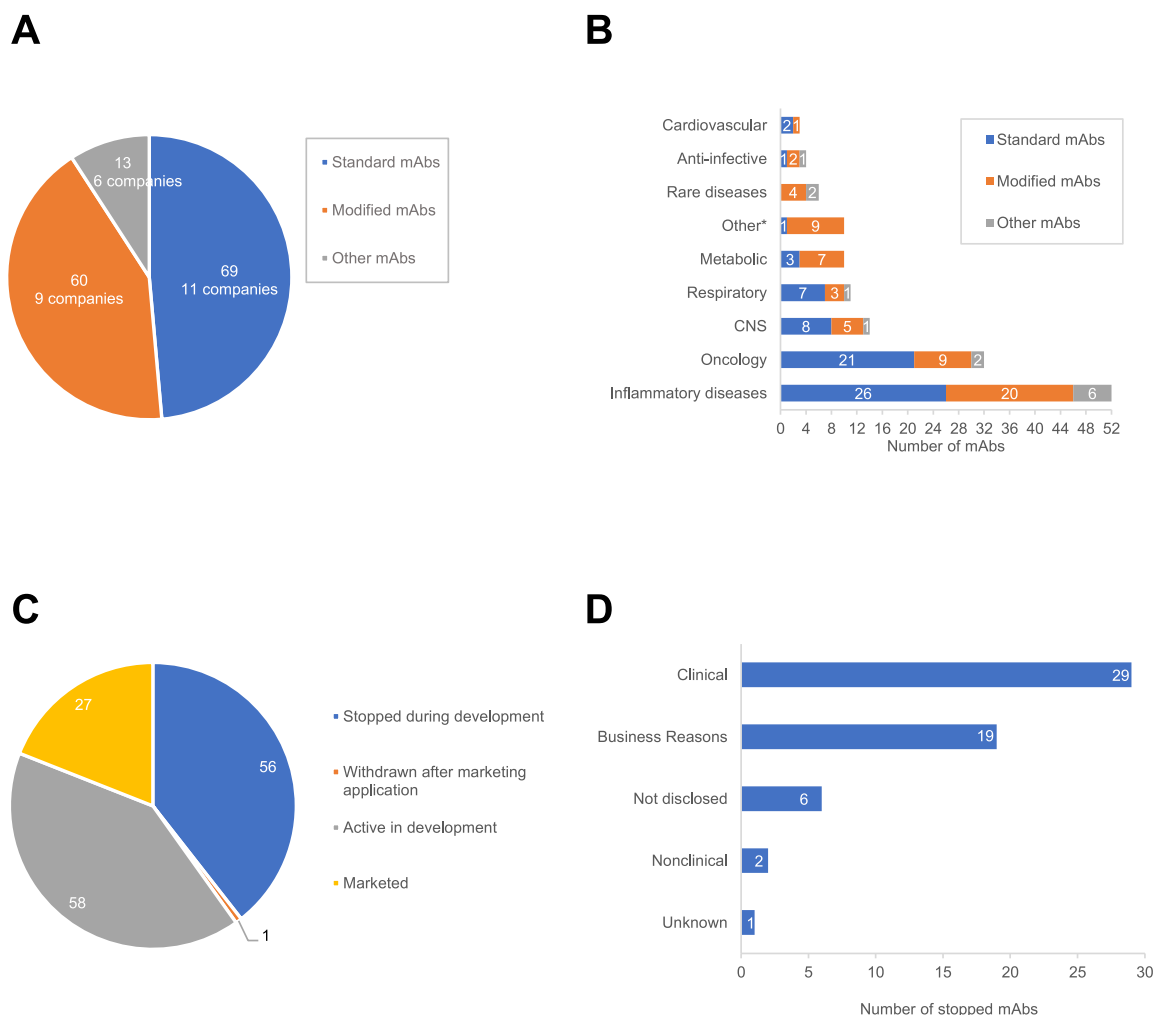
Surveys (data sets) were submitted from 11 different companies, with each company submitting data for five or more mAbs (with a mean of 13 mAb surveys per company). In total, data for 142 mAbs were received: 69 (49%) standard mAbs provided by 11 companies, 60 (42%) modified mAbs provided by 9 companies, and 13 (9%) 'other mAbs' (8 bispecifics, 4 Fabs (antibody fragment) and 1 nanobody) provided by 6

companies (Fig. 2A).

The predominant IgG subtype was IgG1 (55%) followed by IgG4 (32%) and IgG2 (13%) (Supplementary Fig. 1). Seven of the 142 mAbs (5%) were identified as agonists whilst 135/142 (95%) mAbs were identified as antagonists.

Study initiation dates ranged from 1991 to 2019, with 59% of the studies conducted between 1991 and 2011 and 41% of the studies conducted between 2012 and 2019, the significance of this cut-off being the adoption of ICH S6(R1) in 2011.

The mAbs were representative of a range of therapeutic areas, with



**Fig. 2.** Overview of the data sets.

**Fig. 2A.** Modality type and the number of submitting companies: data sets for 142 mAbs were submitted by 11 companies in total (the number of companies providing each mAb type is provided in parenthesis). The majority of the data sets consisted of standard/modified mAbs. **Fig. 2B.** Primary therapeutic areas: The largest number of data sets was intended for inflammatory diseases, followed by oncology, CNS, and respiratory diseases. \*Other therapeutic areas consisted of chronic kidney disease and hematology (2 mAbs each), fibrosis, urology, immunology rare disease, transplant rejection (1 mAb each) and 2 mAbs where the therapeutic area was not disclosed. **Fig. 2C.** Current status: The majority of mAbs were not marketed, with 58/142 (41%) still in active development, and development terminated for 56/142 (39%). **Fig. 2D.** Primary reasons for stopping: Of 56 mAbs for which development was terminated, 28 (50%) programmes were stopped for clinical reasons, and 2 (4%) for nonclinical reasons. For 7 mAbs the reasons for stopping were not disclosed.

inflammatory diseases (52/142, 37%) and oncology (32/142, 23%) being the most highly represented, followed by central nervous system (CNS) (14/142, 10%), respiratory diseases (11/142, 8%) and metabolic diseases (10/142, 7%) (**Fig. 2B**).

The mAbs were either in active development (58/142, 41%), terminated during development (56/142, 39%) (**Fig. 2C**) or marketed (28/142, 20%). One product was subsequently withdrawn post marketing. Of the 56 mAbs terminated, the most common reasons for stopping development were clinical (28 mAbs) or business related (19 mAbs). In 2 cases, development was stopped due to nonclinical findings (**Fig. 2D**).

### 3.2. Data inclusion/exclusion

Data sets provided for 5 mAbs were removed from the analysis, due to only short-term studies or no study data provided (4 mAbs) or the short-term study duration of a single dose with only a one-day observation period (1 mAb). The remaining 137 data sets comprised 121 data sets that included chronic studies (**Fig. 1A**) and 16 data sets with only sub-chronic studies. Surveys containing 'Other mAbs' were treated as an

independent group from standard/modified mAbs due to the low number of case examples provided ( $n = 10$ , as three 'other mAbs' did not meet the initial inclusion criteria). Therefore, the analysis of novel adverse findings identified by chronic toxicity studies comprised 111 data sets for standard/modified mAbs. For an analysis of mAb 'program designs' (section 3.3), all 142 mAbs were evaluated for pharmacological relevance and species use and all data sets with study information ( $n = 141$ ) were used to examine study durations, number of studies within programs, group sizes etc.

### 3.3. mAb program designs

#### 3.3.1. Pharmacological relevance and species use

The surveys included information on the different species tested for pharmacological relevance, the tests used to determine pharmacological relevance for each species, whether each species was subsequently determined to be pharmacologically relevant, and the species ultimately chosen for toxicology studies.

### 3.3.2. Species tested for pharmacological relevance

Sixteen different species were tested for pharmacological relevance across all the submissions (Supplementary Fig. 2a). All but 2 mAb programs tested the cynomolgus monkey for relevance (140 mAb programs), with 7 mAb programs determining cynomolgus monkey was not a pharmacologically relevant species. When tested, the rhesus monkey was always found to be a pharmacologically relevant species (14 mAb programs). Alternative non-rodent species (e.g. minipig, dog and sheep) were tested for a very small number of mAb programs ( $n = 19$ ). The dog was tested by 19 mAb programs and generally found not to be of pharmacological relevance (18 mAb programs). The rabbit was tested for almost a third of the programs (44 mAbs) and was a pharmacologically relevant species in a third of these (15 mAb programs). Approximately three quarters of the programs in the data sets tested the rat and/or mouse (104 and 101 mAb programs, respectively), with approximately 25% of these mAb programs identifying one or both rodent species as pharmacologically relevant.

The range in number of species tested for pharmacological relevance per mAb program was one to 12; the majority (88%) tested two or more species, with the average number of species tested being three (Supplementary Table 3). When only one species was tested (17 mAb programs), this was always an NHP species. When two species were tested (26 mAb programs), this was always an NHP species plus rat or mouse. A high proportion (60%) of mAb programs identified only one relevant species, which was an NHP species for all but 2 mAb programs (in which a TG mouse was the only relevant model).

### 3.3.3. Tests used to determine pharmacological relevance

Four broad testing categories used to determine the pharmacological relevance of each species are shown in Supplementary Table 4. For the mAb programs using cynomolgus monkey (the majority of the data sets), pharmacological relevance was usually determined via multiple tests, with 88% of the mAb programs using two or more tests.

### 3.3.4. Species use in mAb programs

Sixty percent of mAb programs identified only one relevant species (Table 1), yet 77% of mAb programs used only one species in each phase (with the majority using an NHP species only). Both a rodent and non-rodent species were identified as pharmacologically relevant in 28% of mAb programs. However, fewer (23%) mAb programs used multiple (two or more) species within the toxicology program in at least one phase. The main reason that some programs used only the non-rodent species was due to immunogenicity or partial relevance i.e., binding to the rodent homologue but without functional activity (Table 1).

Overall, 96% of mAb programs (136/142 mAbs) used the cynomolgus monkey in one or both phases of the program. There were 4 mAb programs that did not use an NHP species at all, with 3 mAbs using a TG mouse only, and the other using minipig and rat. For the 23% of mAb programs that used multiple species (rodent and non-rodent) for the short-term studies, approximately half retained both for the longer (sub-chronic or chronic) studies whilst the other half reduced to one species (Table 2). More information on this data is included within a separate publication (Prior et al., 2022).

### 3.3.5. Study duration

A high degree of variability was observed in study durations (Fig. 3A). In general, studies with durations of 10 weeks or less were submitted as those supporting FIH trials, whilst studies with durations of 24 weeks or more were submitted as those supporting later development. For the 'sub-chronic' duration of studies (12–16-week duration) a number of scenarios were apparent.

- The shortest study for 26 mAb programs (to support FIH trials).
- An intermediate study for 26 mAb programs (performed either to support FIH or later development, in addition to a short and chronic duration study).

**Table 1**

The number of mAbs, the species identified as pharmacologically relevant, and the species actually used for the toxicity studies (maximal number across the program).

| Pharmacological relevance |                                       | Species used in toxicity studies <sup>a</sup> |                                                                                     |                                                                                    | Additional information                                                                                                                                                                                                                                      |
|---------------------------|---------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| # relevant species        | Rodent/<br>Non-rodent                 | Rodent                                        | Non-rodent                                                                          | Rodent + non-rodent                                                                |                                                                                                                                                                                                                                                             |
| 0                         | n/a: 2 mAbs                           |                                               | C: 1 mAb <sup>1</sup>                                                               | C + R: 1 mAb <sup>2</sup>                                                          | 1 Anti-viral product; C chosen as standard tox model with background data.<br>2 Anti-infective indication, mAb does not bind to any endogenous target; species chosen as standard rodent and non-rodent tox models.                                         |
| 1                         | Rodent: 2 mAbs                        | TG: 1 mAb                                     |                                                                                     | C + TG: 1 mAb <sup>3</sup>                                                         | 3 Target not present in conventional tox species; TG used for pharmacology effects and C for potential side effects (has a related target).                                                                                                                 |
|                           | Non-rodent: 83 mAbs                   | TG: 1 mAb <sup>4</sup>                        | C: 81 mAbs<br>Mt: 1 mAb                                                             |                                                                                    | 4 Chimpanzee identified as relevant, used TG to avoid use of this protected species.                                                                                                                                                                        |
| 2                         | 2 x non-rodents: 15 mAbs              |                                               | C: 10 mAbs<br>Rh: 1 mAb<br>C + Rh: 2 mAbs <sup>5</sup>                              | C + M: 1 mAb <sup>6</sup><br>C + Rh + M: 1 mAb <sup>5,7</sup>                      | 5 Switched non-rodent species for short and long studies.<br>6 Partial relevance (binding but not activity) in mice; included short-term only.                                                                                                              |
|                           | Rodent + non-rodent: 14 mAbs          | TG: 1 mAb <sup>8</sup>                        | C: 5 mAbs <sup>9</sup>                                                              | C + R: 4 mAbs<br>C + M: 3 mAbs<br>C + TG: 1 mAb                                    | 7 Regulatory request to include short mouse study.<br>8 Used TG to avoid use of chimpanzee<br>9 Partial relevance in rodent, C demonstrated higher relevance/binding and/or considered more appropriate species to use.                                     |
| 3 or more                 | All non-rodents: 1 mAb                |                                               | C + Rh: 1 mAb <sup>5</sup>                                                          |                                                                                    | 5 Switched non-rodent species for short and long studies                                                                                                                                                                                                    |
|                           | Rodent (s) + non-rodents (s): 25 mAbs |                                               | C: 2 mAbs <sup>10</sup><br>C + Rh: 1 mAb <sup>5</sup><br>C + Mt: 1 mAb <sup>5</sup> | C + R: 18 mAbs<br>C + GP: 1 mAb<br>MP + R: 1 mAb<br>C + R + M: 1 mAb <sup>11</sup> | 5 Switched non-rodent species for short and long studies<br>10 Immunogenicity or only partial relevance in rodents; NHPs demonstrated higher relevance/binding and/or considered more appropriate species to use.<br>11 Regulatory request to include mouse |

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Table 1 (continued)

| Pharmacological relevance |                   | Species used in toxicity studies <sup>a</sup> |            |                     | Additional information                                  |
|---------------------------|-------------------|-----------------------------------------------|------------|---------------------|---------------------------------------------------------|
| # relevant species        | Rodent/Non-rodent | Rodent                                        | Non-rodent | Rodent + non-rodent |                                                         |
|                           |                   |                                               |            |                     | tox to investigate mortalities in mouse efficacy study. |

M mouse (wild-type), R Rat, TG transgenic mouse; C cynomolgus monkey, GP guinea pig, Mt marmoset, MP minipig, Rh Rhesus monkey.

<sup>a</sup> Highest number of species used (in one or both phases).

Table 2

Combinations of studies within mAb programs.

| Total studies                                                                           | Studies supporting FIH                    | Studies supporting later development | Number of mAbs |
|-----------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------|----------------|
| <b>Programs using non-rodent only in both phases (103 mAbs)</b>                         |                                           |                                      |                |
| 2                                                                                       | 1 short                                   | 1 sub-chronic                        | 12             |
| 2                                                                                       | 1 short                                   | 1 chronic                            | 46             |
| 2                                                                                       | 1 sub-chronic                             | 1 chronic                            | 19             |
| 3                                                                                       | 1 short                                   | 2 chronic                            | 2              |
| 3                                                                                       | 2 short                                   | 1 chronic                            | 2              |
| 3                                                                                       | 1 short + 1 sub-chronic                   | 1 chronic                            | 17             |
| 3                                                                                       | 2 sub-chronic                             | 1 chronic                            | 1              |
| 4                                                                                       | 2 short                                   | 1 sub-chronic + 1 chronic            | 1              |
| 4                                                                                       | 1 short + 1 sub-chronic                   | 2 chronic                            | 1              |
| 4                                                                                       | 1 short + 1 sub-chronic                   | 1 sub-chronic + 1 chronic            | 2              |
| <b>Programs using rodent only in both phases (2 mAbs)</b>                               |                                           |                                      |                |
| 2                                                                                       | 1 short                                   | 1 chronic                            | 1              |
| 3                                                                                       | 2 short                                   | 1 chronic                            | 1              |
| <b>Programs using rodent (R) and non-rodent (NR) in both phases (17 mAbs)</b>           |                                           |                                      |                |
| 4                                                                                       | 1 short (R + NR)                          | 1 chronic (R + NR)                   | 6              |
| 4                                                                                       | 1 sub-chronic (R + NR)                    | 1 chronic (R + NR)                   | 1              |
| 4                                                                                       | 1 short (R + NR)                          | 1 sub-chronic (R + NR)               | 3              |
| 5                                                                                       | 1 short (R) + 2 short (NR)                | 1 sub-chronic (R + NR)               | 1              |
| 5                                                                                       | 1 short (R) + 2 short (NR)                | 1 chronic (R + NR)                   | 2              |
| 5                                                                                       | 1 short (NR) + 1 sub-chronic (R + NR)     | 1 chronic (R + NR)                   | 1              |
| 6                                                                                       | 1 short (R + NR) + 1 sub-chronic (R + NR) | 1 chronic (R + NR)                   | 1              |
| 6                                                                                       | 2 short (R + NR)                          | 1 chronic (R + NR)                   | 1              |
| 8                                                                                       | 2 short (R + NR) + 1 sub-chronic (R + NR) | 1 chronic (R + NR)                   | 1              |
| <b>Programs changing use of rodent (R) and non-rodent (NR) between phases (15 mAbs)</b> |                                           |                                      |                |
| 3                                                                                       | 1 short (R + NR)                          | 1 chronic (NR)                       | 7              |
| 3                                                                                       | 1 short (R + NR)                          | 1 chronic (R)                        | 1              |
| 3                                                                                       | 1 sub-chronic (R + NR)                    | 1 chronic (NR)                       | 1              |
| 3                                                                                       | 1 sub-chronic (R + NR)                    | 1 chronic (R)                        | 2              |
| 4                                                                                       | 1 short (R + NR)                          | 1 sub-chronic + 1 chronic (NR)       | 1              |
| 5                                                                                       | 2 short (NR) + 1 sub-chronic (R + NR)     | 1 chronic (NR)                       | 1              |
| 5                                                                                       | 1 short (R) + 2 short (NR)                | 2 chronic (NR)                       | 1              |
| 5                                                                                       | 2 sub-chronic (NR)                        | 1 chronic (R) + 2 chronic (NR)       | 1 <sup>a</sup> |

5 mAbs from the database were excluded from this analysis as they did not meet initial inclusion criteria (see Fig. 2A).

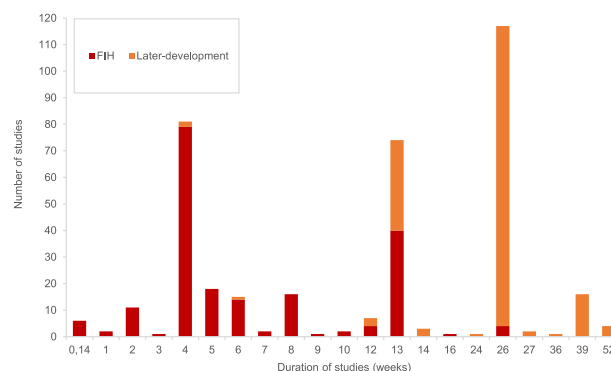
<sup>a</sup> = Unique case where an additional species was included for chronic studies. Short: studies of 10 weeks and less duration; Sub-chronic: studies of 12–16 weeks duration; Chronic: studies of 24 weeks and longer duration.

- The longest study for 16 mAb programs: 9 mAbs for oncology indications following ICH S9; the remaining 7 mAbs were stopped in development (before a chronic study had been performed).

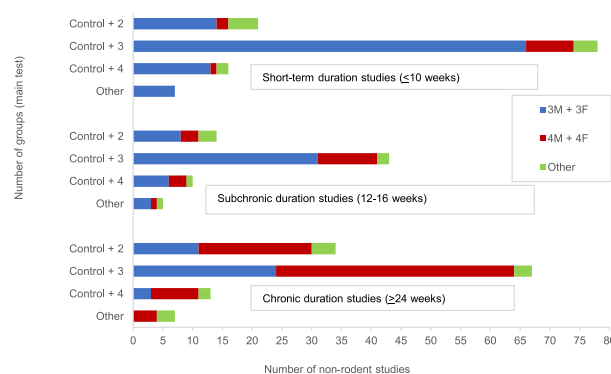
### 3.3.6. Number of studies per mAb program

There was variability in the number of studies performed, with more

A



B



C

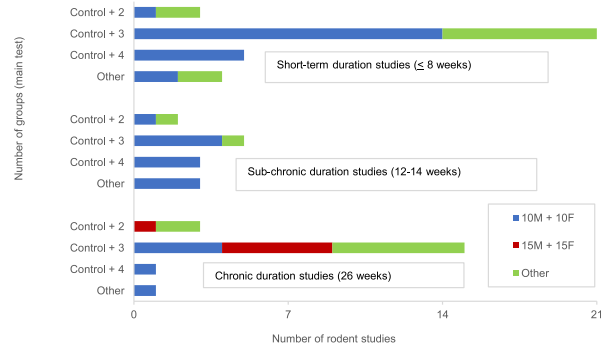


Fig. 3. mAb study designs.

Fig. 3A. Study durations (non-rodent and rodent studies reported in the surveys) in support of FIH and later development. Fig. 3B. Number of groups and animals per group (non-rodents) for the different study duration categories. Other number of groups = control + 1 (1 in each category); control + 5, 6 or 7 (often reflecting interim necropsies for specific dose levels). Other group sizes = 1 sex only; variable sizes per group (e.g., fewer controls, more animals in 1 specific group); 5M + 5F or higher (often reflecting additional purposes/measurements). Fig. 3C. Number of groups and animals per group (rodents) for the different study duration categories. Other number of groups: control + 1 (1 short duration study); control + 5 (1 short and 3 sub-chronic duration studies); control + 6 (1 short and 1 chronic duration study); no control + 3 (1 short duration study). Other group sizes: 5M + 5F (3 short and 1 sub-chronic duration studies); 6M + 6F (2 short duration studies); 12M + 12F (1 sub-chronic duration study); 14M + 14F (3 short and 1 chronic duration studies); 16M (control) + 46M (test) with no females (1 short duration study); 20M + 20F (1 short and 4 chronic duration studies); 22M + 20F (1 chronic duration study); 24M + 24F (TG mice, 1 chronic duration study); 25M + 25F (TG mice, 1 short duration study); 25 M and 25 F (TG mice, low dose group 35M + 35F, 1 chronic duration study).

studies conducted when including two species, as would be expected. The majority (103/137, 75%) of the mAb programs used a single non-rodent species; a high proportion of these (78/103, 76%) conducted one study to support FIH and one study to support later development. For 11 mAb programs, 3 studies were conducted, with one rodent and non-rodent study performed to support FIH, followed by one longer study in only one of these species. For 17 mAb programs, a variety of studies in both rodent and non-rodent species were conducted, ranging from 4 studies (one rodent and non-rodent study to support FIH trial and a second rodent and non-rodent study to support later development) up to 8 studies for one of the mAb programs (Table 2).

### 3.3.7. Group sizes and number of main study groups

Overall, there were 315 non-rodent studies and 66 rodent studies within the data sets of 142 mAbs.

### 3.3.8. Non-rodent studies

For non-rodents, three animals per sex (3 male [M] + 3 female [F]) per group was commonly used for short-term and sub-chronic studies (82% and 67% respectively) and for 31% of chronic studies. A group size of four animals per sex (4M + 4F) per group was commonly used for chronic non-rodent studies (59%) and for 24% of sub-chronic non-rodent studies (Fig. 3B; other group sizes noted in caption).

For all three study duration categories in non-rodents, control plus three test article-dosed study groups were most commonly used, with 64% of short-term, 60% of sub-chronic, and 55% of chronic studies using this design. Control plus two test article-dosed groups was also used in 28% of chronic non-rodent studies.

### 3.3.9. Rodent studies

For rodents, 10 animals per sex (10M + 10F) per group was used for 84% of short-term, 85% of sub-chronic studies and for 30% of chronic studies. A group size of 15 animals per sex or higher was often used in chronic rodent studies (55%) (Fig. 3C; other group sizes noted in caption). Studies with group sizes of 20 or more rodents per sex were occasionally used; this tended to reflect use of mice or when extensive toxicokinetic (TK) or tissue sampling was required in rats.

In rodents, control plus three test article study groups was used for 68% of short-term studies and 75% of chronic studies, though only for 38% of sub-chronic studies.

### 3.3.10. Comparison of the number of animals used between the shortest and longest duration studies

The comparisons were made between paired studies that fell within one of the three study duration categories as follows: short vs. sub-chronic; short vs. chronic; sub-chronic vs. chronic duration studies (section 2.3).

For 77 mAb programs only 2 studies were conducted, allowing a simple comparison. Where a mAb program had performed multiple studies within the same category (e.g., two short duration studies) these almost always used the same number of animals per group. Where a mAb program had performed studies within all three categories, the comparison was made between the shortest and longest performed.

### 3.3.11. Non-rodent studies – overall number of animals used between the shortest and longest duration studies

Some mAb programs reduced the number of groups in the later study, decreasing the overall number of animals used if the group sizes remained the same (16 of 58 mAb programs), or maintaining the same overall number of animals used if the group sizes had increased in the later study (16 of 71 mAb programs). However, most programs used the same number of groups for the two studies, maintaining the same overall number of animals used if the group sizes were the same (36 of 74 mAb programs), or increasing the overall number of animals used if the group sizes had increased in the later study (36 of 74 mAb programs). Other mAb programs increased the number of groups in the later study (21 of

132 mAb programs), increasing the overall number of animals used. When taking both the group sizes and number of groups into consideration, there was an overall increase in non-rodent use for 44% of the mAb programs in the longest duration study, with 40% of mAb programs using the same number of animals and 17% of mAb programs using fewer animals.

### 3.3.12. Non-rodent studies – group size comparisons between the shortest and longest duration studies

There were 132 mAb programs with non-rodent studies in one of the paired categories. Just under half the mAb programs (58/132 mAb programs, 44%) retained the same group sizes between the paired studies; the majority of these mAb programs (45 mAbs) used 3M + 3F in both studies whilst a small proportion (12 mAb programs) used 4M + 4F in both studies. Approximately half the programs (64/132 mAb programs, 48%) increased group sizes from 3M + 3F in the shorter studies to 4M + 4F in the longer studies. There were a small number of mAb programs using variable approaches (e.g., specific groups with higher numbers, lower group sizes for control groups), whilst 3 mAb programs decreased the group size from the shorter to the longer studies (Table 3a).

### 3.3.13. Rodent studies – group size comparisons between the shortest and longest duration studies

Paired rodent studies were available for 23 mAb programs (Table 3b). For 10 mAbs (43%) the group sizes were the same between the paired studies, predominantly using 10M + 10F (9 mAbs). Approximately half the programs (11/23, 48%) increased group sizes from 10M + 10F in the shortest studies to 15M + 15F (6 mAbs) or 20M + 20F (3 mAbs) in the longest studies. There were only two programs (9%) for which the group sizes were reduced in the longest studies (Table 3b).

### 3.3.14. Rodent studies – overall number of animals used between the shortest and longest duration studies

For the majority of rodent programs (15 mAbs, 65%) the number of study groups remained the same in both studies, whilst 1 mAb program increased the number of study groups and 7 mAb programs decreased the number of study groups in the longest studies. Overall, there was an increase in rodent use for 48% of the mAb programs in the longest duration study, with 39% of mAb programs using the same number of animals and 13% of mAb programs using fewer animals.

## 3.4. Novel adverse findings in chronic toxicity studies

For 79/111 (71%) mAb programs, either no toxicities or no new toxicities, were identified in chronic studies. For the other 32/111 (29%) mAb programs, novel adverse findings were observed in chronic studies. Of these 32 programs, novel findings for 17 (15% of 111 total) were not deemed of concern for human safety (Supplementary Fig. 3). Eight of these 17 programs had findings considered to be exaggerated pharmacology or were anticipated based on literature data, previous experience, or were considered a logical progression of toxicities observed in short-term studies. For the other 9/17 programs, the findings were related to ADA formation, which in nonclinical species is usually not predictive to humans as human proteins are recognised by the animal's immune system as "non-self". Likewise, animals are usually not predictive for ADA-mediated adverse reactions to human or humanized therapeutic proteins in humans, including infusion reactions and hypersensitivity reactions (Leach et al., 2014; Rojko et al., 2014; Kronenberg et al., 2017). In one program, the ADA-mediated toxicities could be anticipated based on the findings from the short-term studies in which the molecule was strongly immunogenic. However, there were 15 (13.5% of 111) programs where the novel toxicities were considered of human concern and/or impacted clinical trial design or conduct.

**Table 3a**

Changes in group sizes and number of groups between different non-rodent studies for the same mAb.

| Paired study durations                                | # Animals per group             | # mAbs | # mAbs                          |           |           | Overall animal use** |           |           |
|-------------------------------------------------------|---------------------------------|--------|---------------------------------|-----------|-----------|----------------------|-----------|-----------|
|                                                       |                                 |        | # test groups in longest study* |           |           |                      |           |           |
| Group size the same between shortest & longest study  |                                 |        | Same                            | Increased | Decreased | Same                 | Increased | Decreased |
| Short and sub-chronic study                           | 3M + 3F                         | 10     | 3                               |           | 7         | 3                    |           | 7         |
| Short and chronic study                               | 3M + 3F                         | 27     | 19                              | 3         | 5         | 19                   | 3         | 5         |
| Sub-chronic and chronic study                         | 3M + 3F                         | 8      | 4                               |           | 4         | 4                    |           | 4         |
| Short and sub-chronic study                           | 4M + 4F                         | 1      | 1                               |           |           | 1                    |           |           |
| Short and chronic study                               | 4M + 4F                         | 8      | 5                               | 3         |           | 5                    | 3         |           |
| Sub-chronic and chronic study                         | 4M + 4F                         | 3      | 3                               |           |           | 3                    |           |           |
| Sub-chronic and chronic study                         | Variable <sup>1</sup>           | 1      | 1                               |           |           | 1                    |           |           |
| Group size increased between shortest & longest study |                                 |        |                                 |           |           |                      |           |           |
| Short and chronic study                               | 2M + 2F → 3M + 3F               | 1      |                                 | 1         |           |                      | 1         |           |
| Short and sub-chronic study                           | 3M + 3F → 4M + 4F               | 4      | 2                               | 1         | 1         | 1                    | 3         |           |
| Short and chronic study                               | 3M + 3F → 4M + 4F               | 48     | 27                              | 6         | 15        | 15                   | 33        |           |
| Sub-chronic and chronic study                         | 3M + 3F → 4M + 4F               | 12     | 6                               | 6         |           |                      | 12        |           |
| Short and chronic study                               | 3M + 3F → 5M + 5F               | 2      | 1                               | 1         |           |                      | 2         |           |
| Short and chronic study                               | 3M + 3F → 8M + 8F               | 1      |                                 |           | 1         |                      | 1         |           |
| Short and chronic study                               | Variable → 4M + 5F <sup>2</sup> | 1      |                                 |           | 1         |                      |           | 1         |
| Short and chronic study                               | 4M + 4F → 5M + 5F               | 1      |                                 |           | 1         |                      |           | 1         |
| Short and sub-chronic study                           | 4M + 4F → 5M + 5F               | 1      |                                 |           | 1         |                      |           | 1         |
| Group size decreased between shortest & longest study |                                 |        |                                 |           |           |                      |           |           |
| Short and chronic study                               | 4M + 4F → 3M + 3F               | 2      | 2                               |           |           |                      |           | 2         |
| Sub-chronic and chronic study                         | 5M + 5F → 4M + 4F <sup>3</sup>  | 1      |                                 |           | 1         |                      |           | 1         |

# Number.

\* Compared to the shortest study number of main test groups.

\*\* In the longest study compared with the shortest study.

<sup>1</sup> 4M + 4F (control + high dose groups), 6M + 6F (low dose group) in both the sub-chronic and chronic study.<sup>2</sup> Variable group sizes, 3M+3F (control + high dose group) and 4M + F (low dose group); increased to 4M+5F all groups.<sup>3</sup> Rhesus monkey used for sub-chronic study and cynomolgus monkey used for chronic study.**Table 3b**

Changes in group sizes and number of groups between different rodent studies for the same mAb.

| Paired study durations                                | # Animals per group                | # mAbs | # mAbs                          |           |           | Overall animal use** |           |           |
|-------------------------------------------------------|------------------------------------|--------|---------------------------------|-----------|-----------|----------------------|-----------|-----------|
|                                                       |                                    |        | # test groups in longest study* |           |           |                      |           |           |
| Group size the same between shortest & longest study  |                                    |        | Same                            | Increased | Decreased | Same                 | Increased | Decreased |
| Short and sub-chronic study                           | 10M + 10F                          | 3      | 1                               | 1         | 1         | 1                    | 1         | 1         |
| Short and chronic study                               | 10M + 10F                          | 5      | 5                               |           |           | 5                    |           |           |
| Sub-chronic and chronic study                         | 10M + 10F                          | 1      | 1                               |           |           | 1                    |           |           |
| Short and chronic study                               | 14M + 14F                          | 1      | 1                               |           |           | 1                    |           |           |
| Group size increased between shortest & longest study |                                    |        |                                 |           |           |                      |           |           |
| Short and chronic study                               | 5M + 5F → 25M + 25F <sup>1</sup>   | 1      | 1                               |           |           |                      | 1         |           |
| Short and sub-chronic study                           | 10M + 10F → 15M + 15F              | 1      |                                 |           | 1         |                      | 1         |           |
| Short and chronic study                               | 10M + 10F → 15M + 15F              | 5      | 2                               |           | 3         | 1                    | 4         |           |
| Sub-chronic and chronic study                         | 10M + 10F → 20M + 20F              | 2      | 1                               |           | 1         |                      | 2         |           |
| Short and chronic study                               | 10M + 10F → 20M + 20F              | 1      | 1                               |           |           |                      | 1         |           |
| Short and chronic study                               | 14M + 14F → 20M + 20F <sup>1</sup> | 1      | 1                               |           |           |                      | 1         |           |
| Group size decreased between shortest & longest study |                                    |        |                                 |           |           |                      |           |           |
| Short and chronic study                               | 25M + 25F → 24M + 24F <sup>1</sup> | 1      |                                 |           | 1         |                      |           | 1         |
| Short and sub-chronic study                           | Variable <sup>2</sup> → 12M + 12F  | 1      | 1                               |           |           |                      |           | 1         |

# Number.

\* Compared to the shortest study number of main test groups.

\*\* In the longest study compared with the shortest study.

<sup>1</sup> TG mouse.<sup>2</sup> Variable group sizes: 16M (control) and 46M (two test groups); no females.

### 3.4.1. New toxicities of human concern or impacting clinical trials

In 15 mAb programs for which new toxicities of human concern or impacting clinical trial design or conduct were identified (Fig. 2B) (Supplementary Fig. 3), 12 had findings related to (exaggerated) pharmacology (n = 7) or immune-mediated effects (n = 5). Off-target effects or findings of unknown aetiology were reported in only 3/15 mAb programs or 2.7% of the 111 mAbs analysed (mAb 5, 6, and 15, Table 4).

#### 3.4.1.1. Impact on clinical development of novel adverse toxicities

**3.4.1.1.1. Program termination.** Of the 15 mAb programs with new toxicities observed in chronic toxicity studies, 2 (mAb 14 and 15, Table 4) led to termination of the clinical program. The observed

toxicities were liver toxicity for an IgG4 mAb (mAb 15) and immunotoxicity for an IgG1 mAb (mAb 14). The observed liver toxicity could not be explained based on binding to liver (normal monkey and human liver tissues were negative in an *ex vivo* tissue cross-reactivity study), the mode of action or pharmacology, and the finding was not clinically monitorable as no changes were observed in clinical pathology parameters. The observed immunotoxicity was dose-responsive inflammation in several tissues with a low incidence of ADA and limited immunohistologic granular deposits indicating associated immune complexes (Laffan et al., 2020). *Ex vivo* aggregation data in monkey serum resulting from unexpected high levels of binding of complement factor C1q to this mAb supported a hypothesis that immune complexes of drug aggregates

**Table 4**

The overview of the new toxicities of human concern, the resulting clinical impact, and risk perception.

| mAb           | IgG subtype | Agonist/ antagonist | Findings                                                  | Changes in the clinical study design                                                                                                                                                                                                                              | New toxicities critical? | Comment added by the submitters                                                                                                                                                                            | Indication            |
|---------------|-------------|---------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>mAb 1</b>  | IgG1        | Antagonist          | Common findings for mAbs in NHP (injection site findings) | No changes                                                                                                                                                                                                                                                        | No                       | Injection reactions                                                                                                                                                                                        | Oncology              |
| <b>mAb 2</b>  | IgG1        | Antagonist          | Common findings for mAbs in NHP (ADA-mediated tox)        | <input type="checkbox"/> Increased immunogenicity monitoring                                                                                                                                                                                                      | No                       | ADA effects were not considered of human concern and highest dose was NOAEL.                                                                                                                               | Inflammatory disease  |
| <b>mAb 3</b>  | IgG1        | Antagonist          | Expected exaggerated pharmacology                         | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inform consent form                                                                                                                                                   | No                       | The class effect is described as a potential risk in multiple product labels. Even if the finding was not observed it would have been anticipated in the overall risk assessment for the pediatric trials. | Oncology              |
| <b>mAb 4</b>  | IgG4        | Antagonist          | Pharmacology-related findings                             | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inclusion/exclusion criteria                                                                                                                                          | No                       | Murine ADA formation, principally an anti-human framework response, is different to that observed in human since these were primarily anti-idiotypic in nature.                                            | Inflammatory disease  |
| <b>mAb 5</b>  | IgG1        | Antagonist          | Rare adverse reactions (increases in QTc)                 | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inclusion/exclusion criteria                                                                                                                                          | No                       | No cardiovascular (CV) findings were noted clinically.                                                                                                                                                     | Respiratory disease   |
| <b>mAb 6</b>  | IgG1        | Antagonist          | Serious/rare adverse reactions (off-target effects)       | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inform consent form<br><input type="checkbox"/> Changed stopping criteria                                                                                             | No                       | Exposure multiples were sufficient and considered no risk to humans.                                                                                                                                       | Rare disease          |
| <b>mAb 7</b>  | IgG2        | Antagonist          | Exaggerated pharmacology <sup>a</sup>                     | <input type="checkbox"/> Clinical hold (eventually lifted)                                                                                                                                                                                                        | No                       | Work was done to retrospectively show the effects were not a clinical risk.                                                                                                                                | CNS                   |
| <b>mAb 8</b>  | IgG1        | Antagonist          | Immunotoxicity <sup>b</sup>                               | No changes <sup>b</sup>                                                                                                                                                                                                                                           | Yes                      | Immunostimulation observed in the long-term study instead of expected immunosuppression.                                                                                                                   | Inflammatory disease  |
| <b>mAb 9</b>  | IgG1        | Antagonist          | PD-driven toxicity                                        | <input type="checkbox"/> Increased safety/PD monitoring                                                                                                                                                                                                           | Yes                      | –                                                                                                                                                                                                          | Oncology              |
| <b>mAb 10</b> | IgG4        | Antagonist          | Serious adverse reactions (on-target)                     | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inclusion/exclusion criteria<br><input type="checkbox"/> Changed inform consent form                                                                                  | Yes                      | The range and severity of PD findings greatly exceeded what was expected.                                                                                                                                  | Chronic renal disease |
| <b>mAb 11</b> | IgG1        | Antagonist          | Pharmacology-related findings                             | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inclusion/exclusion criteria<br><input type="checkbox"/> Changed inform consent form<br><input type="checkbox"/> Changed stopping criteria                            | Yes                      | Non-adverse pharmacology related findings in an organ unrelated to the indication resulted in additional specific clinical monitoring.                                                                     | Inflammatory disease  |
| <b>mAb 12</b> | IgG1        | Antagonist          | Common findings for mAbs in NHP (immune complex disease)  | <input type="checkbox"/> Increased immunogenicity monitoring<br><input type="checkbox"/> Changed inclusion/exclusion criteria<br><input type="checkbox"/> Add premedication/rescue medication related to immunogenicity<br><input type="checkbox"/> Clinical hold | Yes                      | NADA were detected in clinical trials. Unknown whether high immunogenicity in humans could result in formation of pathogenic immune complexes.                                                             | Inflammatory disease  |
| <b>mAb 13</b> | IgG2        | Antagonist          | Serious/rare adverse reactions (exaggerated pharmacology) | <input type="checkbox"/> Increased safety/PD monitoring<br><input type="checkbox"/> Changed inform consent form<br><input type="checkbox"/> Changed stopping criteria<br><input type="checkbox"/> Clinical hold <sup>c</sup>                                      | Yes                      | Findings in longer term studies were similar morphologically to shorter term studies, but involved more tissues, and were of greater severity                                                              | Oncology <sup>c</sup> |
| <b>mAb 14</b> | IgG1        | Antagonist          | Serious adverse reactions (immunotoxicities)              | <input type="checkbox"/> Clinical discontinuation                                                                                                                                                                                                                 | Yes                      | Poor correlation with detectable circulating ADA and inflammatory tissue findings resulted in termination of the program due to a potential exaggerated pharmacology risk to humans on repeat dosing.      | Inflammatory disease  |
| <b>mAb 15</b> | IgG4        | Antagonist          | Serious/rare adverse reactions (unknown liver toxicities) | <input type="checkbox"/> Clinical discontinuation                                                                                                                                                                                                                 | Yes                      | Program terminated due to <u>non-monitorable</u> hepatic findings in 6-month toxicity study                                                                                                                | Urology               |

NHP = nonhuman primate, ADA = anti-drug antibodies, NADA = neutralizing anti-drug antibodies, PD = pharmacodynamic, QTc = corrected QT interval. NOAEL = no observed adverse effect.

<sup>a</sup> = The observed exaggerated pharmacology was considered adverse by regulators but not by the company.

<sup>b</sup> = The new toxicities resulted from additional T-cell-dependent antibody response (TDAR) assay which was exclusively included in the long-term studies but not in the short-term study. Thus, the new findings were not certainly related to the duration of treatment. Given that the risk of immune dysregulation was already highlighted in the Inform Consent Form and the guidance to the investigator, the immunotoxicity noted in the long-term study did not lead to changes in the clinical study design.

<sup>c</sup> = There was a clinical hold. The indication was eventually changed from inflammatory disease to oncology and the program was not stopped.

can fix complement, accumulate over time and trigger immune complex disease (Thomson et al., 2021). This mechanism could potentially be more clinically relevant than typical ADA associated immune complex disease in monkeys administered human antibody proteins. Neither of the findings for these 2 mAbs were anticipated based on the FIH-enabling studies and both were considered critical for human risk assessment.

**3.4.1.1.2. Clinical hold.** New toxicities that resulted in temporary clinical hold were identified in 3 of the 15 mAb programs where new toxicities were observed in the chronic studies (mAb 7, 12, and 13, Table 4). For 2 of these mAbs (mAb 7 and 13), new toxicities were considered to be due to exaggerated pharmacology, while the toxicity for the other mAb (mAb 12) was ADA-mediated. These toxicities, or the severity and incidence of the toxicities, were not anticipated based on findings from FIH-enabling studies.

For the mAb program with ADA-mediated toxicity, in the early investigative nonclinical studies high levels of neutralizing ADA were noted at doses lower than those used in the one-month study. This ADA-mediated toxicity initially was not considered to be of human concern as immunogenicity in nonclinical species is not predictive of immunogenicity observed in humans. However, a high level of immunogenicity subsequently observed in the FIH clinical study was considered possibly related to the target. This observation led to modifications in the clinical study design.

For the 2 mAb programs with exaggerated pharmacology, the findings in one program were considered critical for human risk assessment and led to changes in stopping criteria, increased pharmacodynamics (PD)/safety monitoring and changes to the informed consent form. The findings in the second program were not considered a clinical risk and the clinical hold was removed without changes to the clinical protocol and informed consent form.

**3.4.1.1.3. Changes in stopping criteria, inclusion/exclusion criteria, and/or informed consent form.** New toxicities observed for 4/15 programs (mAb 4, 5, 6, and 10, Table 4) were not anticipated, while in 2/15 programs (mAb 3 and 11) new toxicities observed could be anticipated. These new toxicities affected stopping criteria (mAb 6 and 11) and/or inclusion/exclusion criteria (mAb 4, 5, 10, and 11) in clinical trial designs, and/or the informed consent form (mAb 3, 6, 10, and 11). The informed consent form contains safety information regarding investigational new drugs, which allows potential participants to make an informed decision on their participation in the clinical trial. For 2 programs (mAb 10 and 11), findings were considered critical for human risk assessment. One of these had pharmacology-related findings that could have been anticipated based on current literature (mAb 11). The other presented with exaggerated pharmacological findings that were of unanticipated severity and incidence (mAb 10). For 1 program (mAb 3) the exaggerated pharmacology findings were anticipated based on prior experience with similar mAbs.

**3.4.1.1.4. Changes in PD/safety monitoring or immunogenicity monitoring.** For 2/15 mAb programs (mAb 2 and 9, Table 4), it was reported that new toxicities in chronic studies led to increased monitoring in the human trials. For one of these (mAb 9), PD and safety monitoring in humans was implemented due to PD-driven toxicity in the chronic study that was both anticipated from the literature and observed in short-term studies and was considered critical for human risk assessment. For the other program (mAb 2), increased monitoring for immunogenicity was implemented in human trials based on observed ADA-mediated toxicity in the chronic study that was not anticipated but was also not considered critical, given the lack of predictive value for humans.

**3.4.1.1.5. No changes made in human trial design.** New toxicities were reported for 2/15 mAb programs (mAb 1 and 8, Table 4) that were considered of human concern but did not lead to any changes in clinical trial design or conduct. In one of these programs (mAb 1), injection site findings were reported; these were not anticipated but were not considered critical for human risk assessment. In the other program (mAb 8), immunotoxicity, in the form of unexpected immunostimulation, was reported; this finding was not anticipated and was considered critical for human safety. However, given that immune dysregulation was already a highlighted risk in the informed consent form and appropriate clinical monitoring was in place, this finding did not impact clinical trial design.

### 3.5. Increased short-term study duration and rate of new toxicities in the chronic study

For programs that included short-term studies of  $\leq 4$  weeks duration and then conducted chronic studies of  $\geq 24$  weeks duration, the rate of chronic studies producing new toxicities was 7/41 (17%). A lower rate of chronic studies with new toxicities, 3/24 (12.5%), was observed for data sets that included short-term studies of 5–10 weeks duration and  $\geq 24$  weeks chronic duration. For programs that included the FIH-enabling studies of 12–16 weeks and  $\geq 24$  weeks chronic duration, the rate of chronic studies with new toxicities decreased further, to 2/22 (9%) (Fig. 4) (Section 2.3).

## 4. Discussion

For the majority of mAbs in our data sets, a 6-month study provided only limited or no additional safety information vs. FIH-enabling data. Due to the exquisite selectivity of mAbs for their target, unexpected toxicity rarely occurs, observed toxicities are not novel or can be managed in patients enrolled in clinical trials (Brennan et al., 2010). Consequently, limiting the duration of long-term studies in specific cases may be justified, using a weight-of-evidence approach. Additionally, 3Rs approaches within existing regulatory guidance could be further leveraged to reduce and to refine animal use for mAb development.

The high degree of variability in almost every aspect of mAb toxicology programs and study designs observed within these data sets (Table 2) may have been influenced by guidance provided in ICH S6 (R1), different company practice or culture, experience with specific targets or therapeutic indications, and regulatory interactions. Furthermore, various opportunities to reduce the use of animals and incorporate 3Rs principles have been suggested in previous publications (Chapman et al., 2009, 2012). Wider adoption of these recommendations could lead to further optimal use of animals, mutually acceptable to regulators and sponsors. Although the survey did not address whether each mAb had been submitted for regulatory consideration, the fact that chronic toxicity studies were completed implies that regulatory approval for at minimum a FIH-trial had been granted, which would include those mAbs using minimised program designs.

The following common features could be identified from the current data sets.

**Species selection:** Cynomolgus monkey was found to be the only pharmacologically relevant species for most mAbs (57%). When a NHP species was not identified as pharmacologically relevant ( $n = 5$ , Table 1), an alternative non-rodent species (minipig) or a transgenic mouse model was used (20% and 40%, respectively, Table 1). Use of

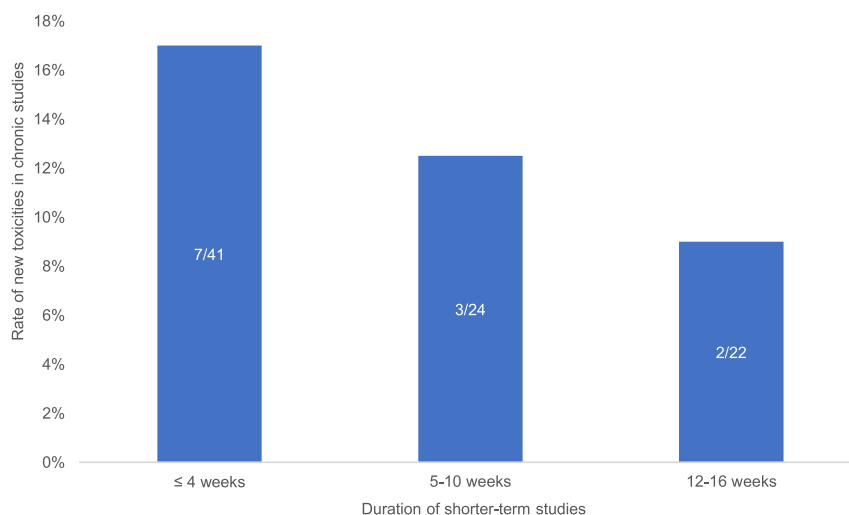


Fig. 4. The rate of chronic studies with new toxicities reduced with increased duration of FIH-enabling study.

alternative non-rodent species requires careful consideration. Currently there is a paucity of published data describing the use of minipig or other non-rodent species for assessing the toxicity of mAbs. Transgenic or knockout animals may be appropriate alternative models, though first need extensive characterisation to understand the pharmacological activity of the model compared to humans and to generate phenotypic background data on the model to allow proper interpretation of safety endpoints. Inclusion of the rat or mouse as a second species, occurred in around a quarter of the data sets, which is similar to previous reports (Prior et al., 2020). Currently, given a shortage of NHPs, FDA encourages the alternative use of other non-rodent species or a relevant rodent species only, where scientifically justified (FDA 2022). This re-enforces the importance of consideration of multiple species, as species other than NHP may be relevant (Namdari et al., 2021; Van der Laan et al., 2010). Where no pharmacologically relevant species is identified and/or the pharmacology can be anticipated, it may be possible to conduct clinical trials without animal use; further guidance is provided in ICH S6(R1).

**Number of species used:** ICH S6(R1) recommends conducting long-term studies in a single species when appropriate, i.e. when toxicity is similar in both species. However, for the mAbs using a rodent and non-rodent species in FIH-enabling studies, approximately half retained both species for the longer studies supporting later development. This occurred irrespective of the study being conducted pre- or post ICH S6(R1) release. For the remaining mAb programs there was a reduction to one species for later studies, progressing the non-rodent in most cases, which has also been observed by others (Prior et al., 2022). While our survey did not determine the rationale for using two species for chronic toxicity studies, the use of two species in half of programs seems high. Applying ICH S6(R1) to reduce chronic toxicity studies to one species consistently would reduce and/or refine animal use, and any deviation to retain the non-rodent species or both non-rodent and rodent species in the chronic study should be appropriately justified.

**Number of studies:** In 35% of mAb programs, toxicity studies were conducted between the FIH-enabling and chronic studies, typically a 13-week study. While not recommended from a regulatory perspective, for some companies an interim study may have provided another opportunity for safety review before the chronic study starts. However, the survey data highlights that three studies of varying durations are unlikely to be more informative than studies of two durations.

**Study duration:** Whilst the most common FIH-enabling study dosing duration was 4 weeks, this ranged from a single dose to 13 weeks. The most common dosing duration supporting later development was 26 weeks, but this ranged from 13 to 52 weeks [4 mAbs, of which one was post-ICH S6(R1)]. The shorter duration FIH-enabling studies often facilitate earlier clinical trial initiation. Thirteen-week duration FIH-enabling studies on the other hand may extend the design and duration of Phase 1 dosing and/or expedite Phase 2 initiation. This duration of study also provides additional data to protect clinical trial participants. If unexpected findings are identified in the 3 month study, this would then inform dose-setting and design for the subsequent chronic toxicity study, if deemed necessary. Choosing a longer study duration as the FIH-enabling study may therefore provide more flexibility in the clinical plan although it is acknowledged this requires additional resource and time early in development. Importantly, there were 22/111 (20%) data sets with only sub-chronic and chronic studies, suggesting that starting with a 13-week FIH-enabling study instead of a 4-week study did not necessarily affect the critical path of their development program. It is acknowledged that for some oncology programs a single 3-month study may support both FIH and longer duration trials; however, these mAbs were outside the scope for the survey.

**Number of groups and group sizes:** Many mAb studies used control plus three test article-dosed groups for both rodent and non-rodent FIH-enabling and chronic toxicity studies, similar to previous reports (Baldrick, 2011; Chapman et al., 2012). Typically, non-rodent chronic studies used fewer groups (control and 2 test article groups) than the FIH-enabling studies (control and 3 test-article groups); however, group sizes often increased such that the same overall number of animals were used. Wider use of fewer test article-dosed groups, particularly for chronic toxicity studies where the dose levels can be chosen from FIH-enabling study results, has been recommended previously (Chapman et al., 2012).

The number of animals per group within a toxicity study is largely a balance between practicality, 3Rs, and having an adequate number to assess trends in results in relation to concurrent controls and historical data. The main test group size commonly employed for non-rodents is 3M + 3 F, as reflected in this data set (Fig. 3B). However, there was a trend for the chronic toxicity studies to use a higher overall number of animals, particularly due to increasing main test group sizes (commonly to 4M + 4F) and/or, less often, the numbers of groups. This is similar to findings found in other mAb data sets (Chapman et al., 2012). Whilst the

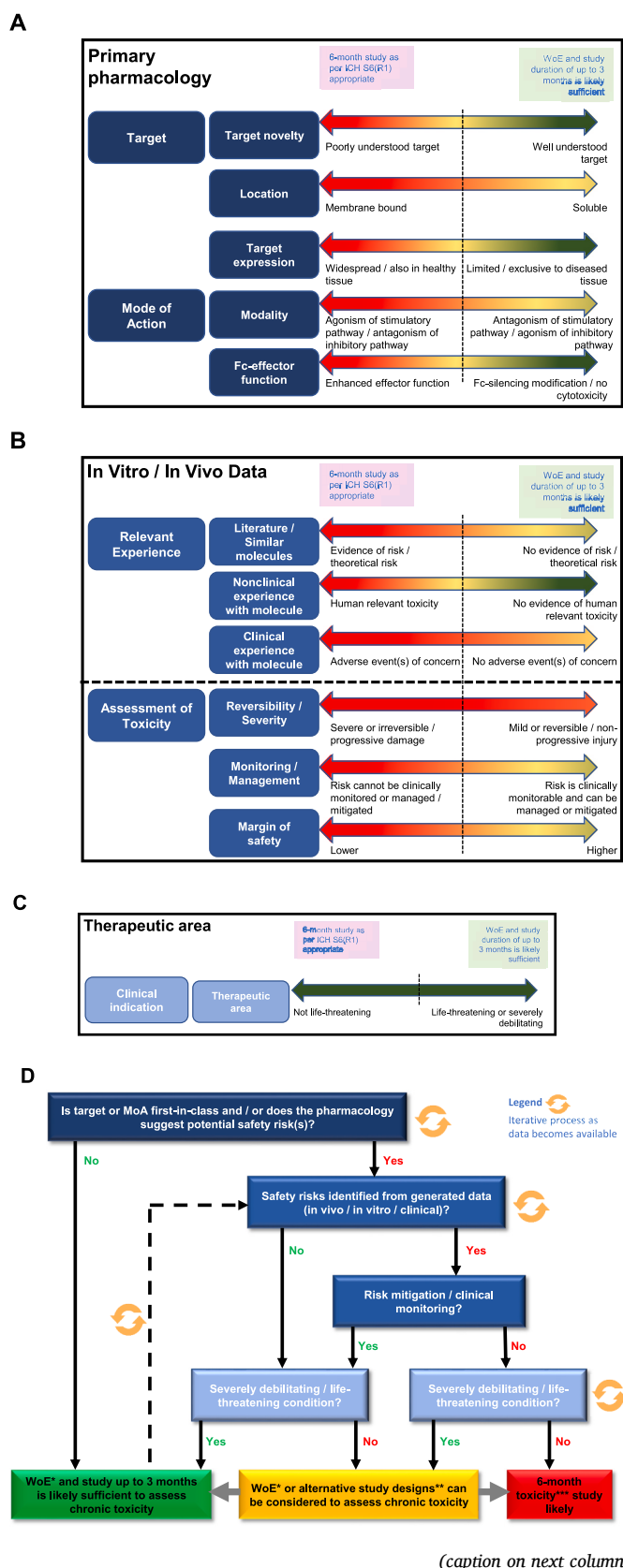


Fig. 5. Integrated weight-of-evidence model.

Fig. 5A. Sliding component section 1: Assessment of target and mechanism of pharmacological action. MoA = mechanism of action. Fig. 5B. Sliding component section 2: Assessment of in vitro/in vivo experience with other similar molecules and the test article itself. Fig. 5C. Sliding component section 3: Consideration of therapeutic area. Fig. 5D. Follow component: Iterative decision model. \* WoE (weight of evidence), including one repeat-dose toxicity study in pharmacologically-relevant species of up to 12 weeks duration, to support clinical development; \*\* examples of alternative study designs include optimized for animal usage, refined dosing duration; \*\*\* optimized for animal usage, recommendations on optimal study designs are included in the discussion.

survey did not explore the reasons for the higher group size in chronic studies, this may have been done to mitigate against potential loss of animals or data during the study due to immunogenicity and/or to have increased interpretative power in the histopathology assessment and/or immunotoxicity endpoints in the “pivotal” study. Working group members have also reported that certain regulatory authorities request to see data from group sizes of 4M + 4F for the chronic studies. One reference to this design can be found in OECD guideline 452 (OECD, 2018) for chronic toxicity which includes the statement that ‘a minimum of 4 per sex per group is recommended’ for non-rodent studies. However, this guideline is typically intended for chemical development after oral administration rather than for injected or orally administered (bio) pharmaceuticals. Nevertheless, a significant proportion of mAb studies are conducted using 3M + 3F non-rodents and these studies have been included within regulatory submissions across multiple regions, underlining this is an acceptable design that should be the starting point for mAb development programs.

Similarly, group sizes increased for chronic rodent studies. The reasons for this increase are likely similar to those for non-rodents, with another common explanation being the use of additional satellite animals for blood sampling. Solutions toward blood sampling for TK purposes have been proposed before, with the additional benefit of being able to compare data directly with toxicity data in the same animal (ICH S3A Q&A, 2017; Li et al., 2019; Williams et al., 2021).

If saturation of pharmacology is typically achieved at the lowest dose and in the absence of a significant reduction in exposure due to ADA, use of two test article-dosed groups is likely sufficient to assess the safety of biopharmaceuticals; a low dose comparable to the predicted clinically efficacious dose and a high dose to achieve systemic exposures 10-fold greater than the maximum predicted clinical exposure. Furthermore, an increase in group size may not be needed in the chronic dosing study, especially where there are no notable sex differences or where there are only expected pharmacology findings in the FIH study.

**Inclusion of recovery animals:** Flexibility around how and when recovery of adverse findings is performed is detailed within ICH S6 (R1) and ICH M3(R2) Q&A guidance [ICH M3(R2) Q&A, 2012]. An evaluation based on scientific assessment alone and well-established knowledge of specific pathophysiology may be sufficient in many cases. If there is a need to monitor reversal of adverse findings, and if the use of animals is warranted for this, ICH S6(R1) provides guidance for inclusion of recovery animals in only one toxicity study within the program (Wange et al., 2021). Our data shows that even when findings were apparent, these were often well-understood. It would therefore seem appropriate to question the inclusion of recovery animals in both studies. A focused analysis of the recovery data from our survey is presented and discussed in a separate publication (Prior et al., 2023).

These and other opportunities to minimize the use of animals were explored in a workshop held with Working Group members and selected regulatory agency representatives in April 2021. One participant presented a single ‘abbreviated’ GLP study design in NHP following ICH S9, to support FIH and registration. The study included one-month and 3-month arms, with reduced group size (2M + 2F) in the control groups and a lower number of dose groups (control + 2) in the one-month arm. By submitting an interim report from the one-month arm to support FIH and the final report for subsequent clinical trials, this alternative approach accelerated clinical development timelines within the advanced cancer setting whilst also reducing animal use by 10 animals compared to a conventional program. While such studies should be considered on a case-by-case basis and are not without some limitations, it is important to note that these options can be implemented in a development strategy and is accepted by regulators. This further highlights the flexibility that currently exists within the regulations.

#### 4.1. Are 6-month studies always required for nonclinical safety assessment of mAbs?

Our assessment showed that in most cases (79/111, 71%) 6-month studies showed toxicities that were not novel; were limited to those mediated by ADA; or could be anticipated based on known pharmacology. These observations confirm that the safety profile in nonclinical toxicity studies of conventional mAbs can be anticipated. Retrospectively, but importantly, the chronic studies did not provide data to better inform clinicians on potential risks in clinical trials in these cases compared with studies of 3 months duration.

Nevertheless, novel adverse findings can occur in chronic toxicity studies. In our data sets, new toxicities that were of human concern or affected clinical development were identified in 13.5% (15/111) of the chronic studies. However, in 7/15 (47%) of these chronic studies, the new toxicities were ultimately not considered critical for human risk assessment by the survey responders. This further highlights that the safety profile of mAbs is relatively benign and can be anticipated or adequately mitigated when compared to small molecules.

In these 7/15 (47%) cases, human relevant toxicities were identified that led to changes in clinical development (see Table 4). In 2 cases (mAb 14 and 15, Table 4), a new toxicity was identified that could not be monitored or mitigated clinically and which subsequently terminated development. One case was an idiosyncratic effect and would not have been detected otherwise; for the second case the mechanism of toxicity was subsequently determined (Laffan et al., 2020). Therefore, a chronic toxicity study is required in some cases to characterize potential harmful effects in humans.

Toxicities observed in the chronic study were often considered to be confirmatory rather than critical for human safety assessment, particularly for findings that were class effects. Whilst useful to confirm the potential for adverse events in clinical trials, the findings were often already identified in sub-chronic studies. An exacerbation in incidence or severity in long-term studies however may further inform clinical safety in some cases, but if findings can already be anticipated or predicted in sub-chronic studies, studies of 6-month duration may not be essential for clinical risk assessment and this strategy should be considered on a case-by-case basis.

By conducting 3-month studies, the rate of chronic studies with new toxicities of human concern was reduced to 9%. This suggests that increasing the duration of FIH-enabling studies to 3 months may be sufficient to identify most if not all clinically-relevant effects. For some mAbs, a 3-month toxicity study is already a standard component of the FIH-enabling study program, as either the only FIH-enabling study or in addition to a shorter (~one-month) study. However, the potential delay to a FIH-trial and the additional test article required for a longer duration study could be perceived to hamper clinical development timelines, particularly when early clinical data is considered to be highly informative. Furthermore, the decision to conduct a 6 month study may only

be determined later in clinical development, potentially delaying overall development timelines. However, the toxicity study is often not on the critical path, especially for FIH study start and the conduct of a single 3-month study could reduce animal use up to 50% with less than two-fold increase in time and resource. Potential limiting factors to conduct of a single 3-month toxicity study can be considered on a case-by-case basis and are often dependent on pharmacological and product development characteristics. For example, opportunistic infections that may emerge following prolonged treatment with an immunosuppressant may require longer observation periods. Rarely, bridging studies may be needed to evaluate process changes, which were introduced after the 3-month study was concluded (e.g., changing cell-lines or novel formulation changes) that cannot be resolved by risk assessment or quality characterisation. In addition, if use of sexually mature NHPs was widely adopted, specifically to assess fertility by histopathological examination in 3 month toxicity studies, this would lead to further pressure on the already short supply of NHPs. However, recent FDA guidance suggests that when fertility parameters can be assessed in rodents, use of sexually mature monkeys may not be required (FDA, 2022).

When considering the design of a single 3-month study, the rodent is recommended if pharmacologically relevant as it is generally sufficiently sensitive to detect toxicity. It is recognised that many sponsors and regulators prefer to use a NHP species and this should be scientifically justified as per ICH S6(R1). As FIH studies frequently use intravenous dosing to achieve maximum exposures whilst the commercial route may be subcutaneous or intramuscular, it would be necessary to include all routes of administration in a single 3-month study. However, adequate coverage of different administration routes can typically be achieved without increasing animal use; groups receiving the test article can be administered drug by one route or another at different dose levels, and it is possible to dose vehicle control animals by both routes if considered necessary.

Taking these considerations into account, for those products where risk can be considered minimal, a single study with a 3-month duration could be sufficient to support later development, similar to the current approach for anti-cancer products. Omitting a 6-month study could provide overall development time savings and would reduce animal use. This warrants further consideration particularly when this results in safe and timely access to innovative medicines for patients. However, whilst the conduct of a single 3-month toxicity study vs. a study of 1- as well as 6-month duration is attractive, it is acknowledged that the probability of novel toxicities not being detected may increase, and in rare cases could introduce harm to patients or volunteers. Not all nonclinical strategies would benefit from a nonclinical approach where a single 3-month FIH-enabling study is conducted that also supports late clinical development for a chronic indication. What is lacking therefore is a methodology that can separate those molecules that are more likely to carry risk from those that are low risk.

#### 4.2. Weight of evidence model

Even though the majority of mAbs in our survey showed no clinically relevant novel toxicity, it is important to determine which molecules would likely benefit from a long-term assessment and which would be amenable to other strategies. Thus, rather than consider a “single 3-month toxicity study” approach for all mAbs, a more scientific approach to mAb development may be achieved by a small shift in the current regulatory position. We propose that an iterative weight-of-evidence (WoE) model can be used for this purpose. The concept of a WoE approach in drug development is not new and is an accepted approach within a number of ICH guidelines (ICH S6(R1), 2011; ICH S11, 2020; ICH S1B(R1), 2022). The WoE approach proposed here is intended to provide the means to determine whether a 6-month study is appropriate for a given mAb and utilises a sliding scale to allow risk assessment of specific risk-factors together with a decision model to drive selection of the optimal dosing duration in animal study/ies

(Fig. 5).

This approach, which has been designed for mAbs with specificity for their target, covers an assessment of the pharmacology of the target (Fig. 5A), the intrinsic properties of the molecule, findings of in vivo animal data (or clinical data if available) for the molecule or similarly acting molecules (Fig. 5B) and the therapeutic area (Fig. 5C). Each factor in the WoE model contains subsections to help guide the assessment but is not all inclusive for every situation and additional factors may be considered based on the individual case (see Supplementary Data 1 for instructions on risk assessment). After risk assessment, three outcomes are suggested in the WoE model, including that WoE and a study up to 3 month duration is sufficient to assess chronic toxicity (3 month approach); WoE and alternative study designs optimized for animal usage and/or refined dosing duration can be considered to assess chronic toxicity (modified approach); 6 month toxicity study likely (6 month approach) (Fig. 5D).

It is essential to note that the entire WoE assessment is iterative. That is, as more data becomes available, this can be fed into the model allowing fine-tuning of the decision or arriving at a new decision with respect to selecting an appropriate duration of the animal study. For example, early in development the existing data informing the overall WoE may suggest a modified approach. This includes designs optimized for animal usage or refined dosing duration, or can include a non-standard study design (e.g., a study containing one-month and 3-month arms, with reduced group size in the control groups and a lower number of dose groups) to be the most optimal approach; however, as new information becomes available, the user can then decide on the need for a 6-month study (see Supplementary Data 2 for case studies). Where there is uncertainty or the risk has increased, a more conservative approach should be considered when completing the WoE model.

Using an iterative approach, multiple case studies with known outcomes were used to guide the development of the WoE model. The final model was then tested using a randomised selection of 5 mAbs from an MEB-database containing all approved mAbs in the EU-region up to 2022 and a further 16 mAbs selected from the current dataset. Of these 21 case studies, 13 mAbs had findings of human concern and 8 mAbs had no novel toxicity identified in their respective long term study. In this exercise the model was considered to correctly anticipate a potential human safety risk in 11 cases (85%). In the remaining two cases, the model suggested a 3 month study, where there was toxicity in the long-term study. This was due to exaggerated pharmacology and therefore was already, in part, anticipated. For 8 mAbs without novel toxicity, the model correctly suggested a 3 month study in 6 cases. Whilst an alternative design/long term study was suggested for the 2 remaining cases, this suggests the model tends towards a more cautious outcome. If we include these 2 cases, the model was considered to correctly advise the development strategy for 90% of the mAbs. Future studies should evaluate the validity of the model more comprehensively using a larger dataset, including a prospective study in which the model is used in parallel with current approaches. This could form the basis for regulatory acceptance of a WoE approach with a robust validation to demonstrate its applicability, comparable to what was done to support the ICH S1B revision (Van der Laan et al., 2016).

The WoE model can facilitate within-company decision making during development of the toxicity testing strategy for a mAb and for planning purposes when determining study and animal requirements. It would also support interactions with regulatory agencies to provide a justification for the choice of studies. The application of a WoE model within regulatory guidance is not novel. The change in risk perception around mAbs, further enhanced by logistical challenges demands a scientific consideration on when longer term studies are not required (FDA, 2022). With this proposed WoE model, which was developed with consideration of WoE approaches described in 2 seminal publications (Leach et al., 2021; ICH S11, 2020), we are for the first time proposing that this approach can also be applied to more conventional toxicology

studies. That is not to say that a WoE approach would always lead to a reduction in longer-term studies. The WoE model was designed to allow for varying risk profiles for each molecule and its application based around scientific discussions both within companies and with regulators early in drug development to establish the validity of such an approach. To achieve wider implementation of this approach, next steps would be to open discussions on the need for 6 month studies, with a view to updating ICH S6(R1).

## 5. Conclusions

The 3Rs opportunities that were identified from our data sets should be more rigorously adopted and are fully in line with both regulatory guidance and previously published recommendations. It should be recognised that mAbs are generally well tolerated, and for the majority of these mAbs novel toxicity is rarely detected in nonclinical studies of longer than 3 months duration. This provides an opportunity to follow a case-by-case approach to reduce animal use compared to the conventional nonclinical toxicology program, by conducting a 3-month study instead of a 6-month study as the longest duration toxicity study. Immediate gains in animal reduction are possible within the scope of ICH S6(R1) in which case-by-case approaches are already permitted. A more science-based WoE approach for mAbs to determine the need for 6-month vs. shorter duration toxicity studies should be employed rather than defaulting to a chronic toxicity study of 6-month duration.

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## CRediT authorship contribution statement

**Hsiao-Tzu Chien:** Data curation, Methodology, Writing – original draft, Writing – review & editing, Visualization. **Helen Prior:** Data curation, Methodology, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Laura Andrews:** Resources, Writing – review & editing. **Leon van Aerts:** Conceptualization, Writing – review & editing. **Annick Cauvin:** Resources, Writing – review & editing. **David O. Clarke:** Conceptualization, Resources, Writing – original draft, Writing – review & editing. **Kaushik Datta:** Resources, Writing – review & editing. **Maggie Dempster:** Resources, Writing – review & editing. **Noel Dybdal:** Resources, Writing – review & editing. **Wendy Freebern:** Resources, Writing – review & editing. **Lolke de Haan:** Conceptualization, Resources, Writing – review & editing, Visualization. **Danuta Herzyk:** Resources, Writing – review & editing. **Adam Hey:** Resources, Writing – review & editing. **Thomas Kissner:** Resources, Writing – review & editing. **Sven Kronenberg:** Resources, Writing – review & editing. **Michael W. Leach:** Conceptualization, Resources, Writing – original draft, Writing – review & editing. **Donna Lee:** Resources, Writing – review & editing. **Katrin Schutte:** Conceptualization, Writing – review & editing. **Fiona Sewell:** Project administration, Writing – review & editing. **Kevin Trouba:** Resources, Writing – review & editing. **Peter Ulrich:** Resources, Writing – review & editing. **Lucinda Weir:** Methodology, Resources, Writing – original draft, Writing – review & editing, Supervision. **Peter van Meer:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

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Additional data regarding the use of one or two species for the mAb program data collected during this project is included in a separate publication (Prior et al., 2022). Likewise, additional data regarding the use of recovery animals within mAb study designs is included in a further separate publication (Prior et al., 2023).

All studies were conducted in accordance with each company's policy on the care, welfare and treatment of laboratory animals and were reviewed by the Institutional Animal Care and Use Committee (IACUC) and/or by an appropriate ethical review process within each company and/or at the institution where the work was performed, if outsourced.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.yrtph.2022.105329>.

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## Further reading

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